

MODULE 1 | PART 3

SET AND FUNCTION THEORY

IOQM & RMO 2026

SET AND FUNCTION THEORY

Set

A set is a collection of well-defined objects that are distinct from each other. Sets are generally denoted by capital letters $A, B, C, \dots$ etc. and the elements of the set by $a, b, c \dots$ etc.

If a is an element of a set A , then we write $a \in A$ and say a belongs to A .

If a does not belong to A then we write $a \notin A$.

Example:

The collection of first five prime natural numbers is a set containing the elements 2, 3, 5, 7, 11.

$\mathbb{Z}^-$ = Set of all negative integers

$$= \{-1, -2, -3, \dots\}$$

$\mathbb{Q}$ = Set of all rational numbers.

$$= \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$$

$\mathbb{R}$ = Set of all real numbers.

$\mathbb{R} - \mathbb{Q}$ = Set of all irrational numbers

Example:

$\sqrt{2}, \sqrt{3}, \sqrt{5}, \dots, \pi, e, \log 2, \dots$ are all irrational numbers.

Some Important Number Sets

$\mathbb{N}$ = Set of all natural numbers

$$= \{1, 2, 3, 4, \dots\}$$

$\mathbb{W}$ = Set of all whole numbers

$$= \{0, 1, 2, 3, \dots\}$$

$\mathbb{Z}$ = Set of all integers

$$= \{\dots - 3, -2, -1, 0, 1, 2, 3, \dots\}$$

$\mathbb{Z}^+$ = Set of all positive integers

$$= \{1, 2, 3, \dots\} = \mathbb{N}.$$

Methods to Write a Set

Roster Method: In this method, a set is described by listing elements, separated by commas and then enclosed by curly brackets

Example:

The set of vowels of English alphabet may be described as $\{a, e, i, o, u\}$.

Set Builder Form: In this case, we write down a property or rule P that gives us all the elements of the set

$$A = \{x : P(x)\}$$

Example:

$$A = \{x : x \in \mathbb{N} \text{ and } x = 2n \text{ for } n \in \mathbb{N}\}$$

$$A = \{2, 4, 6, \dots\}$$

Example:

$$B = \{x^2 : x \in \mathbb{Z}\}$$

$$\text{i.e., } B = \{0, 1, 4, 9, \dots\}$$

Cardinal Number of a Set

When a set is finite, the number of elements in a set is called the cardinal number of a set. The cardinal number of a set A is denoted by $n(A)$ or $|A|$.



1. Describe the following set by set property method $\{0, 3, 8, 15, 24, 35\}$.



2. Write the following set by roster method:

The set of all natural numbers ' x ' such that $4x + 9 < 50$.



Types of Sets

Null Set or Empty Set

A set having no element in it is called an empty set or a null set or a void set. It is denoted by ϕ or $\{\}$.

Example:

$$A = \{x \in \mathbb{N} : 5 < x < 6\} = \phi$$

A set consisting of at least one element is called a non-empty set or a non-void set.

Singleton

A set consisting of a single element is called a singleton set.

Example:

Then set $\{0\}$ is a singleton set

Finite Set

A set that has only finite number of elements is called a finite set.

Example:

$$A = \{a, b, c, d, e\}$$

Order of a Finite Set

The number of elements in a finite set is called the order of the set A and is denoted by $|A|$ or $n(A)$. It is also called cardinal number of the set.

Example:

$$A = \{a, b, c, d\} \Rightarrow n(A) = 4$$

Infinite Set

A set that has an infinite number of elements is called an infinite set.

Example:

$$\mathbb{N} = \{1, 2, 3, 4, \dots\} \text{ is an infinite set.}$$

Equal Sets

Two sets A and B are said to be equal if every element of A is an element of B and every element of B is an element of A .

If sets A and B are equal, we write $A = B$ and if A and B are not equal then we write $A \neq B$.

Example:

$$A = \{1, 2, 6, 7\} \text{ and } B = \{6, 1, 2, 7\}$$

$$\Rightarrow A = B$$



Equivalent Sets

Two finite sets A and B are equivalent if they have the same number of elements.

$$\text{i.e. } n(A) = n(B)$$

Example:

$$A = \{1, 3, 5, 7\}, B = \{a, b, c, d\}$$

$$n(A) = 4 \text{ and } n(B) = 4 \Rightarrow n(A) = n(B)$$

Note:

Equal sets are always equivalent but equivalent sets may not be equal.

Subsets

Let A and B be two sets. If every element of A is an element of B , then A is called a subset of B . If A is a subset of B , we write $A \subseteq B$.

Example:

$$A = \{1, 2, 3, 4\} \text{ and } B = \{1, 2, 3, 4, 5, 6, 7\}$$

$$\Rightarrow A \subseteq B$$

Proper Subset

If A is a subset of B and $A \neq B$ then A is a proper subset of B , and we write $A \subset B$.



Note:

- ❖ Every set is a subset of itself.
i.e., $A \subseteq A$ for all A
- ❖ Empty set ϕ is a subset of every set.
- ❖ Clearly $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.
- ❖ The total number of subsets of a finite set containing n elements is 2^n .

Universal Set

A set consisting of all possible elements that occur in the discussion is called a universal set and is denoted by U .

Example:

If $A = \{1, 2, 3\}$, $B = \{2, 4, 5, 6\}$,
 $C = \{1, 3, 5, 7\}$, then $U = \{1, 2, 3, 4, 5, 6, 7\}$
can be taken as the universal set.

Note:

All sets in the discussion are contained in the universal set of that discussion.

Power Set

Let A be any set. The set of all subsets of A is called power set of A and is denoted by $P(A)$.

Example:

Let $A = \{1, 2\}$ then
 $P(A) = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$

Example:

Let $P(\phi) = \{\phi\}$

$$\Rightarrow P(P(\phi)) = \{\phi, \{\phi\}\}$$

$$\Rightarrow P(P(P(\phi))) = \{\phi, \{\phi\}, \{\{\phi\}\}, \{\phi, \{\phi\}\}\}$$

(Please look this notation very carefully as students often get confused)

Note:

- ❖ If $A = \phi$ then $P(A)$ has only one element.
- ❖ Power set of a given set is always non empty.



3. Two finite sets A and B have m and n elements respectively. If the total number of subsets of A is 112 more than the total number of subsets of B , then the value of m is

- (A) 7 (B) 9
(C) 10 (D) 12

[KEAM 2009]





4. The set $A = \{x : x \in \mathbb{R}, x^2 = 16 \text{ and } 2x = 6\}$ equals

- (A) ϕ (B) $\{14, 3, 4\}$
(C) $\{3\}$ (D) $\{4\}$



5. If $A = \{2, 3\}$, find the value of $n(P(P(P(A))))$.



6. If $A = \{x, y\}$, then the power set of A is:

- (A) $\{x^y, y^x\}$
(B) $\{\phi, x, y\}$
(C) $\{\phi, \{x\}, \{2y\}\}$
(D) $\{\phi, \{x\}, \{y\}, \{x, y\}\}$



7. Let S be a set of real numbers with mean M . If the mean of the sets $S \cup \{15\}$ and $S \cup \{15, 1\}$ are $M + 2$ and $M + 1$, respectively then how many elements does S have?

[PRMO 2014]



Some Operation on Sets

Union of Two Sets

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Example:

$$A = \{1, 2, 3\}, B = \{2, 3, 4\} \text{ then}$$

$$A \cup B = \{1, 2, 3, 4\}$$

Intersection of Two Sets

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Example:

$$A = \{1, 2, 3\}, B = \{2, 4, 5\} \text{ then}$$

$$A \cap B = \{2\}$$

Difference of Two Sets

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

Example:

$$A = \{1, 2, 3\}, B = \{2, 3, 4\}; A - B = \{1\}$$

Complement of a Set

$$A' \text{ or } A^c = \{x : x \notin A \text{ but } x \in U\} = U - A$$

Example:

$$U = \{1, 2, \dots, 10\}, A = \{1, 2, 3, 4, 5\} \text{ then}$$

$$A' = \{6, 7, 8, 9, 10\}$$

De-Morgan's Laws

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

$$A - (B \cup C) = (A - B) \cap (A - C)$$

$$A - (B \cap C) = (A - B) \cup (A - C)$$

Distributive Laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Commutative Laws

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associative Laws

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Note:

- ❖ $A \cap \phi = \phi$
- ❖ $A \cap U = A$
- ❖ $A \cup \phi = A$
- ❖ $A \cup U = U$
- ❖ $A \cap B \subseteq A$
- ❖ $A \cap B \subseteq B$
- ❖ $A \subseteq A \cup B$
- ❖ $B \subseteq A \cup B$
- ❖ $A \subseteq B \Rightarrow A \cap B = A$
- ❖ $A \subseteq B \Rightarrow A \cup B = B$
- ❖ $(A')' = A$
- ❖ $A \subseteq B \Leftrightarrow B' \subseteq A'$

Disjoint Sets

If $A \cap B = \phi$, then A, B are said to be disjoint.

Example:

$$\text{If } A = \{1, 2, 3\}, B = \{7, 8, 9\}, \text{ then}$$

$$A \cap B = \phi$$

Note:

$$A \cap A' = \phi$$

$\therefore A, A'$ are disjoint.

Symmetric Difference of Sets

$$A \Delta B = (A - B) \cup (B - A)$$

Properties Related to Difference

If A and B are any two sets, then

- ❖ $A - B = A \cap B'$
- ❖ $B - A = B \cap A'$
- ❖ $A - B = A \Leftrightarrow A \cap B = \phi$
- ❖ $(A - B) \cup B = A \cup B$
- ❖ $(A - B) \cap B = \phi$
- ❖ $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

Open Interval and Closed Interval

Interval Type	Notation	Includes Endpoints?	Example Set-Form
Open	(a, b)	No	$\{x : a < x < b\}$
Closed	$[a, b]$	Yes	$\{x : a \leq x \leq b\}$
Half-open	$[a, b)$ or $(a, b]$	One endpoint	$\{x : a \leq x < b\}$, etc.
Infinite types	(a, ∞) , $(-\infty, b]$	-	Includes $> a$ or $\leq b$



8. If X and Y are two sets and X' denotes the complement of X , then $X \cap (X \cup Y)'$ is equal to

- (A) X (B) Y
 (C) ϕ (D) $X \cap Y$

[NCERT EP]



9. Let $P = \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$ and $Q = \{\theta : \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$ be two sets. Then

- (A) $P \subset Q$ and $Q - P \neq \phi$
 (B) $Q \not\subset P$
 (C) $P \not\subset Q$
 (D) $P = Q$

[IIT JEE 2011]



10. Let $A = \{x : x \in \mathbb{R}, |x| < 1\}$;

$B = \{x : x \in \mathbb{R}, |x - 1| \geq 1\}$ and

$A \cup B = \mathbb{R} - D$, then the set D is:

- (A) $\{x : 1 < x \leq 2\}$
 (B) $\{x : 1 \leq x < 2\}$
 (C) $\{x : 1 \leq x \leq 2\}$
 (D) none of these



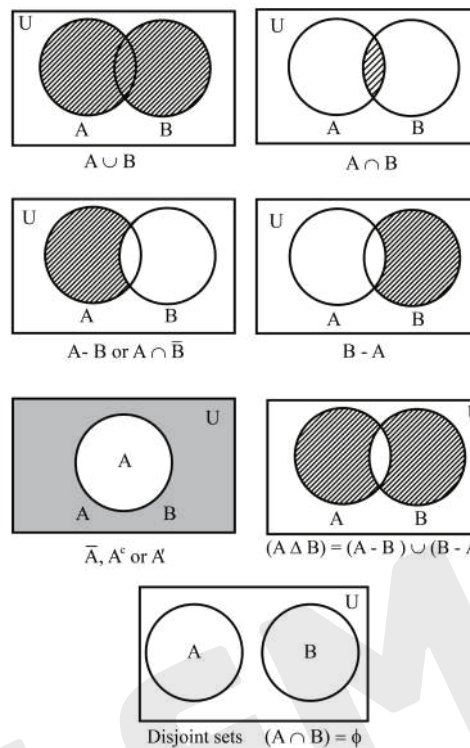
11. If $aN = \{ax : x \in \mathbb{N}\}$, then the set $6N \cap 8N$ is equal to:

- (A) $8N$ (B) $48N$
(C) $12N$ (D) $24N$



Venn Diagrams

Venn diagrams are powerful tools to represent a set operation. In these diagrams, the universal set is represented as a rectangle and subsets of the universal sets are represented by circles of smaller radius such that they completely lie inside the rectangle. In the diagrams, the common area between any two figures represents elements that are common in the two sets corresponding to those figures. In the figures below, shaded region is representative of set operations written below the figures.

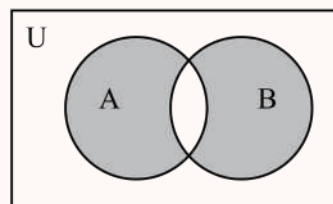


Note:

$$A \cap A' = \phi, A \cup A' = U$$



12. The shaded region in the figure represents



- (A) $A \cap B$
(B) $A \cup B$
(C) $(A - B) \cup (B - A)$
(D) $A - B$

[KEAM 2010]



13. Let $X = \{1, 2, 3, \dots, 10\}$ and $A = \{1, 2, 3, 4, 5\}$. Then the number of subsets B of X such that $A - B = \{4\}$ is

- (A) 2^5 (B) 2^4
(C) $2^5 - 1$ (D) 1

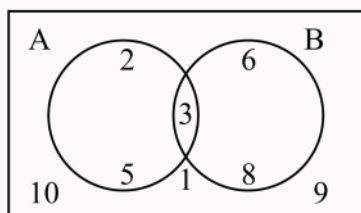
[KEAM 2013]



15. A and B are two sets such that $n(A - B) = 14 + x$, $n(B - A) = 3x$ and $n(A \cap B) = x$. Draw a Venn diagram to illustrate this information and if $n(A) = n(B)$ then find the value of x .



14. From the adjoining Venn diagram, determine the following sets:



- (i) $A \cup B$ (ii) $A \cap B$
(iii) $A - B$ (iv) $(A \cap B)'$



16. Let $X = \{1, 2, 3, 4, 5\}$. The number of different ordered pairs (Y, Z) that can be formed such that $Y \subseteq X, Z \subseteq X$ and $Y \cap Z$ is empty is

- (A) 2^5 (B) 5^3
(C) 5^2 (D) 3^5

[AIEEE 2012]



17. In a group of 50 persons, 14 drink tea but not coffee and 30 drink tea. If each person drinks either tea or coffee, then find:

- (i) How many drink tea and coffee both?
- (ii) How many drink coffee but not tea?



Some Important Results on Number of Elements in Sets

Let A, B and C be finite sets, and U be the finite universal set, then

- ❖ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- ❖ $n(A \cup B) = n(A) + n(B)$
 $\Leftrightarrow A, B$ are disjoint sets.
- ❖ $n(A - B) = n(A) - n(A \cap B)$
i.e., $n(A - B) + n(A \cap B) = n(A)$

❖ $n(A \Delta B) = \text{No. of elements that belong to exactly one of } A \text{ or } B$

$$= n((A - B) \cup (B - A))$$

$$= n(A - B) + n(B - A)$$

$[(A - B) \text{ and } (B - A) \text{ are disjoint}]$

$$= n(A) - n(A \cap B) + n(B) - n(A \cap B)$$

$$= n(A) + n(B) - 2 \times n(A \cap B)$$

❖ $n(A \cup B \cup C) = n(A) + n(B) + n(C)$
 $- n(A \cap B) - n(B \cap C) - n(A \cap C)$
 $+ n(A \cap B \cap C)$

❖ Number of elements in exactly two of the sets A, B, C

$$= n(A \cap B) + n(B \cap C) + n(C \cap A) - 3 \times n(A \cap B \cap C)$$

❖ Number of elements in exactly one of the sets A, B, C

$$= n(A) + n(B) + n(C) - 2 \times n(A \cap B) - 2 \times n(B \cap C) - 2 \times n(A \cap C) + 3 \times n(A \cap B \cap C)$$

❖ $n(A' \cup B') = n((A \cap B)')$
 $= n(U) - n(A \cap B)$

❖ $n(A' \cap B') = n((A \cup B)')$
 $= n(U) - n(A \cup B)$

❖ $n(A_1 \cup A_2 \cup \dots \cup A_n)$
 $= \sum n(A_i) - \sum_{i \neq j} n(A_i \cap A_j)$
 $+ \sum_{i \neq j \neq k} n(A_i \cap A_j \cap A_k)$
 $+ \dots + (-1)^{n-1} n(A_1 \cap A_2 \cap \dots \cap A_n)$



18. *In a group of 40 students, 26 take tea, 18 take coffee and 8 take neither of the two. How many take both tea and coffee.



20. In a certain town 25% families own a cell phone, 15% families own a scooter and 65% families own neither a cell phone nor a scooter. If 1500 families own both a cell phone and a scooter, then the total number of families in the town is

- (A) 10,000 (B) 20,000
(C) 30,000 (D) 40,000

[KEAM 2008]



19. Find the number of n digit numbers formed using first 5 natural numbers, which contain the digits 2 and 4 essentially.



21. *A market research group conducted a survey of 1000 consumers and reported that 720 consumers like product A and 450 consumers like product B , what is the least number that must have liked both products?





22. In a town of 10,000 families it was found that 40% families buy newspaper A , 20% families buy newspaper B , 10% families buy newspaper C , 5% families buy A and B , 3% buy B and C and 4% buy A and C . If 2% families buy all the three newspapers. Find the number of families which buy none of A, B and C .

[NCERT EP]



23. *There are 20 students participating in an after-school program offering classes in yoga, bridge, and painting. Each student must take at least one of these three classes, but may take two or all three. There are 10 students taking yoga, 13 taking bridge, and 9 taking painting. There are 9 students taking at least two classes. How many students are taking all three classes?



24. In a study about a pandemic, data of 900 persons was collected. It was found that
190 persons had symptom of fever,
220 persons had symptom of cough,
220 persons had symptom of breathing problem,
330 persons had symptom of fever or cough or both,
350 persons had symptom of cough or breathing problem or both,
340 persons had symptom of fever or breathing problem or both,
30 persons had all three symptoms (fever, cough and breathing problem).
If a person is chosen randomly from these 900 persons, then the probability that the person has at most one symptom is

[JEE Advanced 2022]





25. Let $S = \{1, 2, 3, \dots, 9\}$.

For $k = 1, 2, \dots, 5$, let N_k be the number of subsets of S , each containing five elements out of which exactly k are odd. Then

$$N_1 + N_2 + N_3 + N_4 + N_5 =$$

[JEE Advanced 2017]



26. *Let S be a set with six elements. In how many different ways can one select two not necessarily distinct subsets of S so that the union of the two subsets is S ? The order of selection does not matter; for example, the pair of subsets $\{a, c\}, \{b, c, d, e, f\}$ represents the same selection as the pair $\{b, c, d, e, f\}, \{a, c\}$.

[AIME]



27. Define an ordered triple (A, B, C) of sets to be minimally intersecting if

$$|A \cap B| = |B \cap C| = |C \cap A| = 1$$

and $A \cap B \cap C = 0$.

For example, $(\{1, 2\}, \{2, 3\}, \{1, 3, 4\})$ is a minimally intersecting triple. Let N be the number of minimally intersecting ordered triples of sets for which each set is a subset of $\{1, 2, 3, 4, 5, 6, 7\}$. Find the remainder when N is divided by 1000.

Note: $|S|$ represents the number of elements in the set S .



28. What is the number of ordered pairs (A, B) where A and B are subsets of $\{1, 2, \dots, 5\}$ such that neither $A \subseteq B$ nor $B \subseteq A$?

[PRMO 2014]



Introduction to Relations

Cartesian Product of Two Sets

If A and B are two non-empty sets, then the set of all ordered pairs (a, b) such that $a \in A, b \in B$ is called the Cartesian product of A and B , and is denoted by $A \times B$. Thus, $A \times B = \{(a, b) : a \in A, b \in B\}$ (read as A cross B).

Note:

- ❖ Two ordered pairs are equal, if and only if the corresponding first elements are equal and the second elements are also equal,
i.e., $(x, y) = (u, v)$ iff $x = u, y = v$.
- ❖ $A \times A \times A = \{(a, b, c) : a, b, c \in A\}$.
Here (a, b, c) is called an ordered triplet.
- ❖ The Cartesian product $\mathbb{R} \times \mathbb{R}$ represents the set $\mathbb{R} \times \mathbb{R} = \{(x, y) : x, y \in \mathbb{R}\}$ which represents the coordinates of all the points in two dimensional space.
- ❖ The Cartesian product $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ represents the set $\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) : x, y, z \in \mathbb{R}\}$ which represents the coordinates of all the points in three-dimensional space.



Relation

- ❖ A relation R from a non-empty set A to a non-empty set B is a subset of the Cartesian product $A \times B$. The subset is derived by describing a relationship between the first element and the second element of the ordered pairs in $A \times B$. The second element is called the image of the first element.
- ❖ A relation may be represented algebraically either by the **roster method** or by the **set-builder method**.
- ❖ An **arrow diagram** is a visual representation of a relation.

Domain, Range and Codomain

- ❖ The set of all first elements of the ordered pairs in a relation R from a set A to a set B is called the **domain** of the relation R .
- ❖ The set of all second elements in a relation R from a set A to a set B is called the **range** of the relation R .
- ❖ The whole set B is called the **codomain** of the relation R .
Note that $\text{range} \subset \text{codomain}$.



Inverse of a Relation

Let A, B be two sets and let R be a relation from A to B . Then, the inverse of R , denoted by R^{-1} , is a relation from B to A obtained from R by interchanging the components of ordered pairs in R . Thus,

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

Clearly, $\text{Domain}(R) = \text{Range}(R^{-1})$ and $\text{Range}(R) = \text{Domain}(R^{-1})$.

Relations Defined on a Set A

Relations defined from a non-empty set A to itself are called relations defined "in set A " or "on set A ".

- ❖ A relation R in a set A is called an empty relation, if no element of A is related to any element of A , i.e., $R = \phi \subset A \times A$.
- ❖ A relation R in a set A is called universal relation, if each element of A is related to every element of A , i.e., $R = A \times A$.
- ❖ Both the empty relation and the universal relation are some times called trivial relations.



29. Find x and y , if $(x - y, x + y) = (6, 10)$ are

- (A) $x = 6, y = 10$ (B) $x = 2, y = 8$
(C) $x = 4, y = 4$ (D) $x = 8, y = 2$

[NCERT EP]



30. If $A \times B = \{(p, q), (p, r), (m, q), (m, r)\}$, then

- (A) $A = \{p, q\}, B = \{m, r\}$
(B) $A = \{m, q\}, B = \{p, r\}$
(C) $A = \{m, r\}, B = \{p, q\}$
(D) $A = \{p, m\}, B = \{q, r\}$

[NCERT TB]



31. *The domain of the relation R given by $R = \left\{ (x, y) : y = x + \frac{6}{x}; \text{ where } x, y \in \mathbb{N} \text{ and } x < 6 \right\}$ is

- (A) $\{1, 2, 3, 4, 5\}$ (B) $\{1, 2, 3, 4\}$
(C) $\{1, 2, 3\}$ (D) $\{1, 2\}$

[NCERT EP]



Simple Counting Problems

- ❖ If $n(A) = p$ and $n(B) = q$, then $n(A \times B) = p \times q$
- ❖ If $n(A) = p, n(B) = q$, then the total number of possible relations from the set A to set B is 2^{pq}
- ❖ For a set A with $n(A) = m$, the number of relations defined on set A is 2^{m^2}
Of these, the empty set ϕ and $A \times A$ are two extreme cases.



32. If $n(A) = 4, n(B) = 3, n(A \times B \times C) = 24$ then, $n(C) =$

- (A) 288 (B) 1
(C) 12 (D) 2

[KEAM 2005]



33. *If $A = \{x : x \in \mathbb{W}, x < 2\}$

$B = \{x : x \in \mathbb{N}, 1 < x < 5\}, C = \{3, 5\},$

then the number of elements in $A \times (B \cup C)$ is

[NCERT EP]



Introduction to Functions

A relation f from a set A to a set B is said to be a function if every element of set A has one and only one image in set B .

In other words, a function f is a relation from a non-empty set A to a non-empty set B such that the domain of f is A and no two distinct ordered pairs in f have the same first element.

Note:

If f is a function from A to B and $(a, b) \in f$, then $f(a) = b$, where b is called the image of a under f and a is called the pre-image of b under f .

The function f from A to B is denoted by $f : A \rightarrow B$.



34. If $f(x) = ax + b$, where a and b are integers, $f(-1) = -5$ and $f(3) = 3$, then a and b are equal to

- (A) $a = -3, b = -1$ (B) $a = 2, b = -3$
(C) $a = 0, b = 2$ (D) $a = 2, b = 3$

[NCERT EP]



36. The range of the function $f : X \rightarrow \mathbb{R}$, $f(x) = x^3 + 1$, where $X = \{-1, 0, 3, 9, 7\}$ is

- (A) $\{0, 1, 28, 344, 730\}$
(B) $\{0, 730\}$
(C) $\{1, 28, 344, 730\}$
(D) $\{-1, 0, 3, 9, 7\}$



35. Let $f(x) = \frac{\alpha x^2}{x+1}$, $x \neq -1$. The value of α for which $f(a) = a$, ($a \neq 0$) is

- (A) $1 - \frac{1}{a}$ (B) $\frac{1}{a}$
(C) $1 + \frac{1}{a}$ (D) $\frac{1}{a} - 1$

[KEAM 2010]



37. *Find the number of maps

$f : \{1, 2, 3\} \rightarrow \{1, 2, 3, 4, 5\}$ such that $f(i) \leq f(j)$ whenever $i < j$.

[IOQM 2022]



Algebra of Functions



38. If f and g are two real valued functions defined as $f(x) = 2x + 1, g(x) = x^2 + 1$, then the incorrect statement is

- (A) $(f + g)(x) = x^2 + 2x + 2$
- (B) $(f - g)(x) = 2x - x^2$
- (C) $fg(x) = 2x^3 + x^2 + 2x + 2$
- (D) $\frac{f}{g}(x) = \frac{2x + 1}{x^2 + 1}$

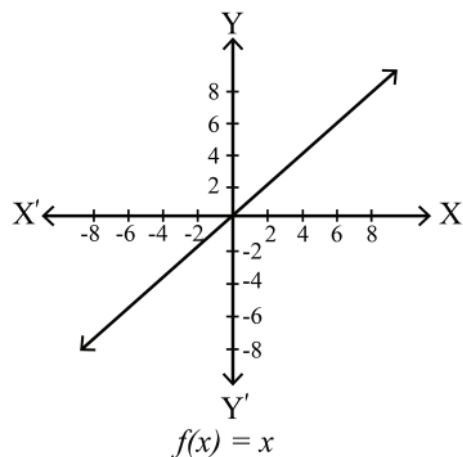
[NCERT EP]



Simple Real Functions

Identity Function

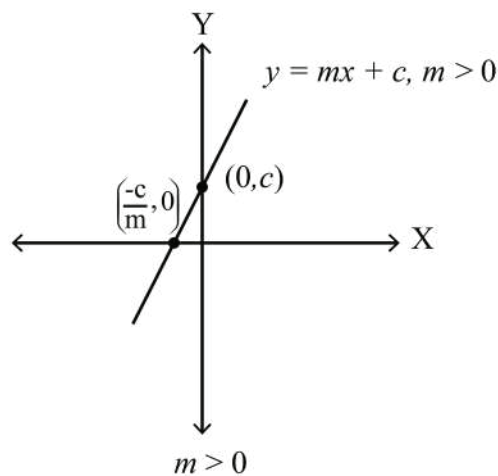
Let $\mathbb{R}$ be the set of real numbers. Define a real valued function $f : \mathbb{R} \rightarrow \mathbb{R}$ by $y = f(x) = x$ for each $x \in \mathbb{R}$. This function is called the identity function. Here the domain and range of f are $\mathbb{R}$. The graph is a straight line passing through the origin as shown in figure.

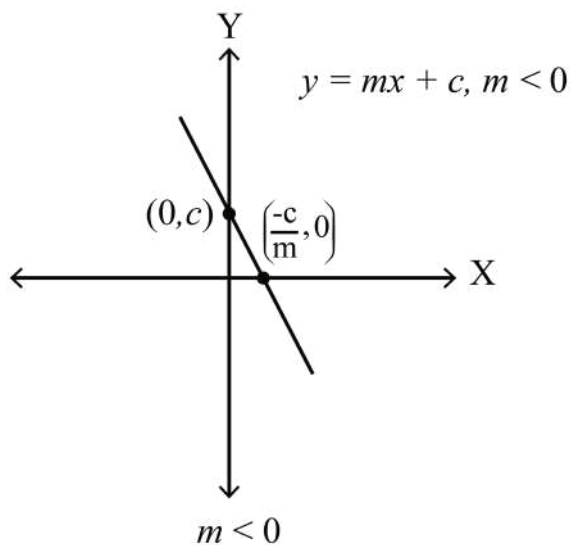


Linear Function

A linear function, $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = mx + c, m \neq 0$

- ❖ Domain: $\mathbb{R}$
- ❖ Range: $\mathbb{R}$
- ❖ Linear function is strictly increasing on $\mathbb{R}$ when $m > 0$ and is strictly decreasing on $\mathbb{R}$ when $m < 0$
- ❖ $f(x) = mx$ satisfies the functional equation $f(x + y) = f(x) + f(y)$





39. Let $\phi(x) = \frac{b(x-a)}{b-a} + \frac{a(x-b)}{a-b}$, where $x \in \mathbb{R}$ and a and b are fixed real numbers with $a \neq b$. Then $\phi(a+b)$ is equal to

- (A) $\phi(ab)$ (B) $\phi(-ab)$
 (C) $\phi(a) + \phi(b)$ (D) $\phi(a-b)$

[KEAM 2012]

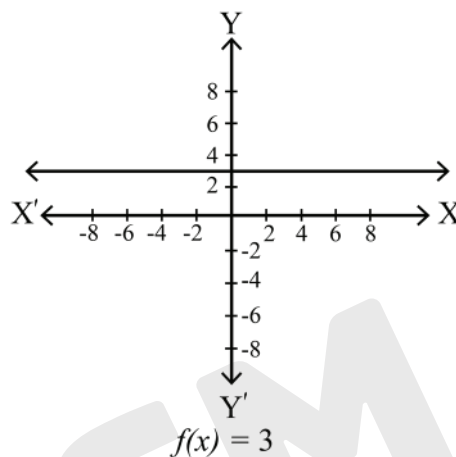


Constant Function

Define the function $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$y = f(x) = c, \forall x \in \mathbb{R}$ where c is a constant.

Here domain of f is $\mathbb{R}$ and its range is $\{c\}$.

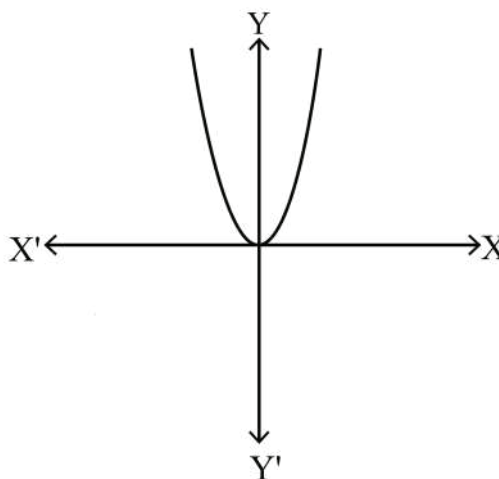


Square Function

Square function $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by

$$f(x) = x^2$$

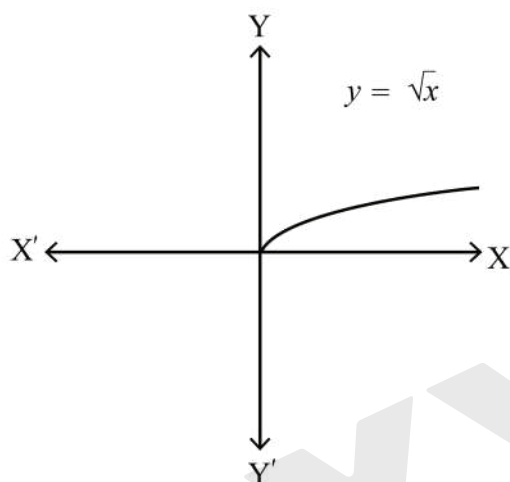
- ❖ Domain: $\mathbb{R}$
- ❖ Range: $[0, \infty)$
- ❖ Square function decreases in $(-\infty, 0)$ and increases in $(0, \infty)$



Square Root Function

Square root function, $f : [0, \infty) \rightarrow \mathbb{R}$ is given by $f(x) = \sqrt{x}$

- ❖ Domain: $[0, \infty)$
- ❖ Range: $[0, \infty)$
- ❖ Square root function increases throughout its domain $[0, \infty)$



40. The domain of the function f given by $f(x) = \sqrt{x-1}$ is

- (A) $(-\infty, \infty)$ (B) $(1, \infty)$
(C) $[1, \infty)$ (D) $[0, \infty)$

[KEAM 2020]



41. The equation $\sqrt{3x^2 + x + 5} = x - 3$, where x is real, has

- (A) no solution
(B) exactly one solution
(C) exactly two solutions
(D) exactly four solutions

[JEE Main 2014]



Quadratic Functions

$f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = ax^2 + bx + c$, $a \neq 0$ is the general quadratic function

Note:

- ❖ The graph of $y = f(x)$ is a parabola. This parabola can be obtained using completion of squares.
- ❖ The vertex of the parabola is given by $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$
- ❖ The parabola opens upwards if $a > 0$ and opens downwards if $a < 0$.

❖ The roots α and β of the corresponding quadratic equation $ax^2 + bx + c = 0$ is obtained by finding the points of intersection of $y = ax^2 + bx + c$ and $y = 0$, i.e., the x -axis.

❖ The range is $\left[-\frac{D}{4a}, \infty\right)$ if $a > 0$ and $\left(-\infty, -\frac{D}{4a}\right]$ if $a < 0$.



42. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = x^2 + 9$. The range of f is

- (A) $\mathbb{R}$
- (B) $[3, \infty)$
- (C) $[9, \infty)$
- (D) $(-\infty, -9] \cup [9, \infty)$

[KEAM 2023]



43. If $f(x) = 2x^2 + bx + c$ and $f(0) = 3$ and $f(2) = 1$, then $f(1)$ is equal to

- (A) 1
- (B) 2
- (C) 0
- (D) -2

[KEAM 2007]



44. The range of the function

$$f(x) = x^2 + 2x + 2 \text{ is}$$

- (A) $(1, \infty)$
- (B) $(2, \infty)$
- (C) $(0, \infty)$
- (D) $[1, \infty)$

[KEAM 2014]



Polynomial Functions

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be a polynomial function if for each $x \in \mathbb{R}$,
 $y = f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$,
where n is a non-negative integer and $a_0, a_1, a_2, \dots, a_n \in \mathbb{R}, a_n \neq 0$.



45. The equations $x^5 + ax + 1 = 0$ and $x^6 + ax^2 + 1 = 0$ have a common root. Then a is equal to

- (A) 0 (B) -2
(C) -3 (D) -1

[KEAM 2017]

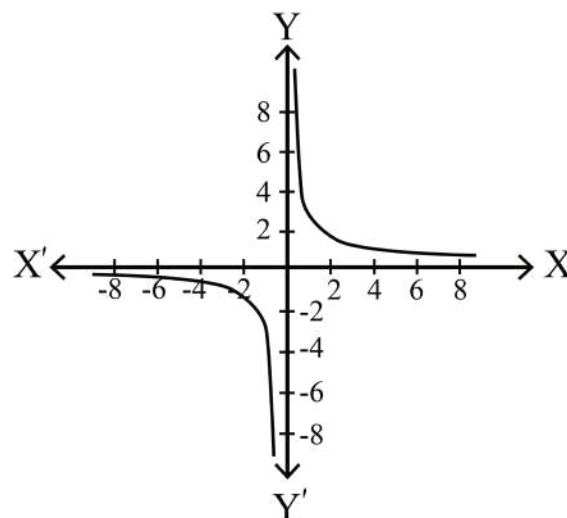


Reciprocal Function

$f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ given by $f(x) = \frac{1}{x}$

Domain: $\mathbb{R} - \{0\}$

Range: $\mathbb{R} - \{0\}$



46. Range of $f(x) = \frac{1}{1 - 2 \cos x}$ is

- (A) $\left[\frac{1}{3}, 1\right]$
(B) $\left[-1, \frac{1}{3}\right]$
(C) $(-\infty, -1] \cup \left[\frac{1}{3}, \infty\right)$
(D) $\left[-\frac{1}{3}, 1\right]$

[NCERT EP]



Rational Functions

Rational functions are functions of the type $\frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are polynomial functions of x defined in a domain where $g(x) \neq 0$.



47. The range of the function $f(x) = \frac{x^2 + 8}{x^2 + 4}$, $x \in \mathbb{R}$ is

- (A) $\left[-1, \frac{3}{2}\right]$ (B) $(1, 2]$
(C) $(1, 2)$ (D) $[1, 2]$

[KEAM 2012]



48. If x is a real number, then $\frac{x}{x^2 - 5x + 9}$ must lie between

- (A) $\frac{1}{11}$ and 1 (B) -1 and $\frac{1}{11}$
(C) -11 and 1 (D) $-\frac{1}{11}$ and 1

[KEAM 2015]



Piecewise Defined Functions



49. Let

$$f(x) = \begin{cases} 1 - 5x, & \text{if } x < -2 \\ x^2 - 2x, & \text{if } -2 \leq x \leq 1. \\ -1 + 2x, & \text{if } x > 1 \end{cases}$$

Then the value of $f(-1)$ is equal to

- (A) -3 (B) 3
(C) -1 (D) 1

[KEAM 2020]



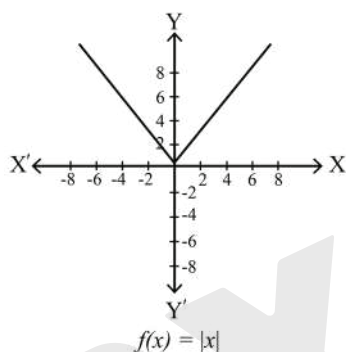
Modulus Function

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = |x|$ for each $x \in \mathbb{R}$ is called the modulus function. For each non-negative value of x , $f(x)$ is equal to x . But for negative values of x , the value of $f(x)$ is the negative of the value of x .

$$\text{i.e., } f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Domain: $\mathbb{R}$

Range: $[0, \infty)$



50. The range of the function

$$f(x) = 1 - |x - 2| \text{ is}$$

- (A) $(-\infty, 1]$ (B) $(-\infty, 1)$
(C) $[1, \infty)$ (D) $(1, \infty)$

[NCERT EP]



51. The domain of $f(x) = \frac{1}{\sqrt{x + |x|}}$ is

- (A) $[0, \infty)$ (B) $(0, \infty)$
(C) $\mathbb{R}$ (D) $(-\infty, 0)$

[NCERT EP]



Modulus Equations



52. The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is

- (A) 2 (B) 1
(C) 3 (D) 4

[JEE Main 2021]

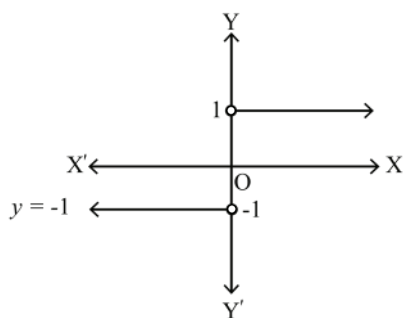


Signum Function

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$$

is called the signum function. The domain of the signum function is $\mathbb{R}$ and the range is the set $\{-1, 0, 1\}$.



53. The range of the function given by $\frac{|x-4|}{x-4}$ is

- (A) $\{1\}$ (B) $\{-1\}$
(C) $\{1, -1\}$ (D) $\{1, -1, 0\}$

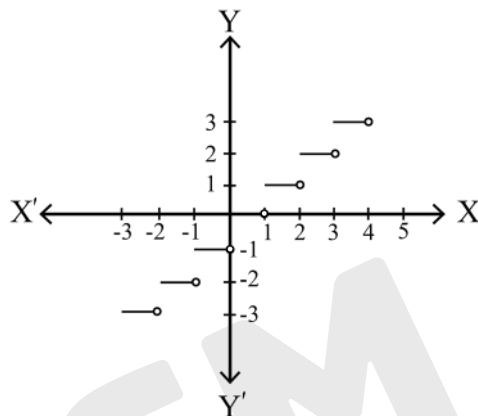
[NCERT EP]



Greatest Integer Function

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

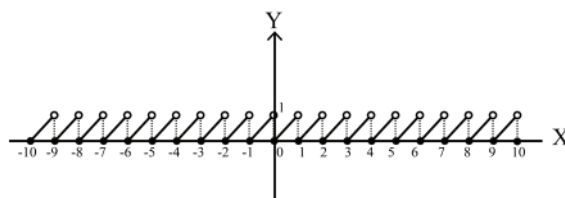
$f(x) = [x], x \in \mathbb{R}$ assumes the value of the greatest integer less than or equal to x . Such a function is called the greatest integer function.



Fractional Part Function

$f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \{x\} = x - [x]$

- ❖ Domain: $\mathbb{R}$
- ❖ Range: $[0, 1)$



54. If $[x]^2 - 5[x] + 6 = 0$, where $[.]$ denote the greatest integer function, then

- (A) $x \in [3, 4]$ (B) $x \in (2, 3]$
(C) $x \in [2, 3]$ (D) $x \in [2, 4)$

[NCERT EP]



Types of Functions

One-One Functions

A function $f : X \rightarrow Y$ is defined to be one-one (or injective), if the images of distinct elements of X under f are distinct,

$$\text{i.e., } x_1, x_2 \in X, f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

Onto Functions

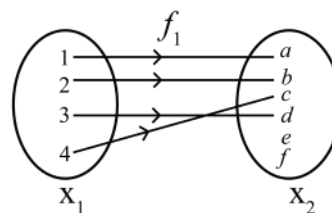
A function $f : X \rightarrow Y$ is defined to be onto (or surjective), if every element of Y is the image of some element of X under f .

i.e., for every $y \in Y$ there exists an element $x \in X$ such that $f(x) = y$.

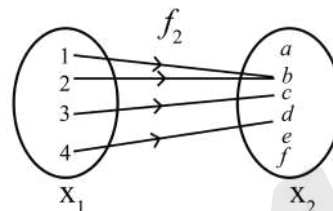
Bijjective Functions

A function $f : X \rightarrow Y$ is said to be one-one and onto (or bijective), if f is both one-one and onto.

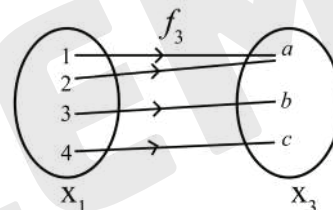
Example:



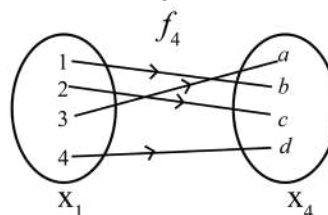
Injective



Neither injective nor surjective



Surjective



Bijjective



55. Let $\mathbb{N}$ be the set of natural numbers and the function $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(n) = 2n + 3, \forall n \in \mathbb{N}$. Then f is

- (A) surjective (B) injective
(C) bijective (D) none of these

[NCERT EP]



56. Let $A = \{0, 1\}$ and $\mathbb{N}$ be the set of natural numbers. Then the mapping $f : \mathbb{N} \rightarrow A$ defined by $f(2n - 1) = 0$, $f(2n) = 1$, $\forall n \in \mathbb{N}$, is

- (A) one-one and onto
- (B) onto but not one-one
- (C) one-one but not onto
- (D) neither one-one nor onto

[NCERT EP]



57. The function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x - 5 \left[\frac{x}{5} \right]$, where $\mathbb{N}$ is the set of natural numbers and $[x]$ denotes the greatest integer less than or equal to x , is

- (A) one-one and onto
- (B) onto but not one-one
- (C) neither one-one nor onto
- (D) one-one but not onto

[JEE Main 2017]



58. Which of the following functions from $\mathbb{Z}$ into $\mathbb{Z}$ are bijections?

- (A) $f(x) = x^3$
- (B) $f(x) = x + 2$
- (C) $f(x) = 2x + 1$
- (D) $f(x) = x^2 + 1$

[NCERT EP]





59. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 7 - 3x$ is

- (A) not one-one (B) not onto
(C) one-one and onto (D) even

[KEAM 2021]



60. *Let $A = [-1, 1]$. Then, for the following functions defined on A , the number of onto functions is

- (i) $f(x) = \frac{x}{2}$ (ii) $g(x) = |x|$
(iii) $h(x) = x|x|$ (iv) $k(x) = x^2$

[NCERT EP]



61. *Let f be a one-to-one function from the set of natural numbers to itself such that $f(mn) = f(m)f(n)$ for all natural numbers m and n . What is the least possible value of $f(999)$?

[PRMO 2014]



Composition of Functions

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be two functions. Then the composition of f and g , denoted by $g \circ f$, is defined as the function $g \circ f : A \rightarrow C$ given by

$$g \circ f(x) = g(f(x)), \forall x \in A$$



Note:

- ❖ If $f : A \rightarrow B$ and $g : B \rightarrow C$ are one-one, then $g \circ f : A \rightarrow C$ is also one-one.
- ❖ If $f : A \rightarrow B$ and $g : B \rightarrow C$ are onto, then $g \circ f : A \rightarrow C$ is also onto.
- ❖ Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be the given functions such that $g \circ f$ is one-one. Then f is one-one.
- ❖ Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be the given functions such that $g \circ f$ is onto. Then g is onto.
- ❖ If $f : X \rightarrow Y$, $g : Y \rightarrow Z$ and $h : Z \rightarrow S$ are functions, then $h \circ (g \circ f) = (h \circ g) \circ f$.



62. Let $f = \{(1,2), (3,5), (4,1)\}$ and $g = \{(2,3), (5,1), (1,3)\}$. Then the CORRECT statement(s) among the following IS/ARE

- (A) $g \circ f = \{(2,5), (5,2), (1,5)\}$
- (B) $f \circ g = \{(2,5), (5,2), (1,5)\}$
- (C) $g \circ f = \{(1,3), (3,1), (4,3)\}$
- (D) $f \circ g = \{(1,3), (3,1), (4,3)\}$

[NCERT EP]



63. Let $f(x) = x^2$ and $g(x) = \sqrt{9+x}$. Then the value of $(f \circ g - g \circ f)(4)$ is equal to

- (A) 6
- (B) $\sqrt{6}$
- (C) $\sqrt{8}$
- (D) 8

[KEAM 2021]



64. If $f(x) = 3x + 5$ and $g(x) = x^2 - 1$, then $(f \circ g)(x^2 - 1)$ is equal to

- (A) $3x^4 - 3x + 5$
- (B) $3x^4 - 6x^2 + 5$
- (C) $6x^4 + 3x^2 + 5$
- (D) $6x^4 - 6x + 5$

[KEAM 2016]



65. If $f(x) = x + 1$ and $g(x) = 2x$, then $f(g(x))$ is equal to

- (A) $2(x + 1)$ (B) $2x(x + 1)$
(C) x (D) $2x + 1$

[KEAM 2012]



66. If $f(x) = \sqrt{x}$ and $g(x) = 2x - 3$, then domain of $(f \circ g)(x)$ is

- (A) $(-\infty, -3)$ (B) $(-\infty, -\frac{3}{2})$
(C) $[-\frac{3}{2}, 0]$ (D) $[\frac{3}{2}, \infty)$

[KEAM 2014]



67. Let $f(x) = \frac{x-1}{x+1}$, $x \in \mathbb{R} - \{0, -1, 1\}$.

If $f^{n+1}(x) = f(f^n(x))$, $\forall n \in \mathbb{N}$, then

$f^6(6) + f^7(7)$ is equal to

- (A) $\frac{7}{6}$ (B) $-\frac{3}{2}$
(C) $\frac{7}{12}$ (D) $-\frac{11}{12}$

[JEE Main 2022]



68. If $g(x) = x^2 + x - 1$ and

$(g \circ f)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to

- (A) $-\frac{3}{2}$ (B) $-\frac{1}{2}$
(C) $\frac{1}{2}$ (D) $\frac{3}{2}$

[JEE Main 2020]





69. If $f(x) = ax + b$ and $g(x) = cx + d$, then $f[g(x)] - g[f(x)]$ is equivalent to

- (A) $f(a) - g(c)$ (B) $f(c) + g(a)$
 (C) $f(d) - g(b)$ (D) $f(b) - g(b)$

[KEAM 2008]



70. Let $f_1(x) = \frac{2}{3} - \frac{3}{3x+1}$, and for $n \geq 2$, define $f_n(x) = f_1(f_{n-1}(x))$. The value of x that satisfies $f_{1001}(x) = x - 3$ can be expressed in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

[AIME]



Inverse of a Function

- ❖ A function $f : X \rightarrow Y$ is defined to be invertible, if there exists a function $g : Y \rightarrow X$ such that $g \circ f = I_X$ and $f \circ g = I_Y$, where I_X and I_Y are identity functions defined on X and Y respectively. The function g is called the inverse of f and is denoted by f^{-1} .
- ❖ A function $f : X \rightarrow Y$ is invertible if and only if f is a bijective function.
- ❖ Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two invertible functions. Then $g \circ f$ is also invertible with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.



71. Let $f : \mathbb{N} \rightarrow Y$ be a function defined as $f(x) = 4x + 3$ where

$$Y = \{y \in \mathbb{N} : y = 4x + 3 \text{ for some } x \in \mathbb{N}\}.$$

Then the inverse of f is

- (A) $g(y) = \frac{y-3}{4}$ (B) $g(y) = \frac{3y+4}{3}$
 (C) $g(y) = 4 + \frac{y+3}{4}$ (D) $g(y) = \frac{y+3}{4}$

[NCERT TB, AIEEE 2008]





- 72.** If $A = \{a, b, c, d\}$ and the function $f = \{(a, b), (b, d), (c, a), (d, c)\}$, then f^{-1} is
- (A) $f^{-1} = \{(a, b), (b, d), (c, a), (d, c)\}$
 (B) $f^{-1} = \{(b, a), (d, b), (a, c), (c, d)\}$
 (C) $f^{-1} = \{(a, d), (d, c), (c, a), (c, d)\}$
 (D) f^{-1} does not exist

[NCERT EP]



- 74.** If $f\left(\frac{x+1}{2x-1}\right) = 2x$, $x \in \mathbb{N}$, then the value of $f(2)$ is equal to
- (A) 1 (B) 4
 (C) 3 (D) 2

[KEAM 2016]



- 73.** Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x - 4$. Then $f^{-1}(x)$ is given by
- (A) $\frac{x+4}{3}$ (B) $\frac{x}{3} - 4$
 (C) $3x + 4$ (D) None of these

[NCERT EP]



- 75.** Let the function f be defined by $f(x) = \frac{2x+1}{1-3x}$. Then $f^{-1}(x)$ is
- (A) $\frac{x-1}{3x+2}$ (B) $\frac{3x+2}{x-1}$
 (C) $\frac{x+1}{3x-2}$ (D) $\frac{2x+1}{1-3x}$

[KEAM 2002]



Special Functions

Special Function I

$$f(x) = \frac{ax+b}{cx-a}$$

$$\Rightarrow f(f(x)) = x$$

$$\Rightarrow f(x) = f^{-1}(x)$$



76. If $f(x) = \frac{x+1}{x-1}$, then the value of $f(f(x))$ is equal to

(A) x (B) 0

(C) $-x$ (D) 1

[KEAM 2017]



77. Let $f : \mathbb{R} - \left\{\frac{\alpha}{6}\right\} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{5x+3}{6x-\alpha}$. Then the value of α for which $(f \circ f)(x) = x$, for all $x \in \mathbb{R} - \left\{\frac{\alpha}{6}\right\}$, is

- (A) 6 (B) 8
(C) 5 (D) No such α exists.

[JEE Main 2021]



78. If $f(x) = \frac{\alpha x}{x+1}$, $x \neq -1$, for what value of α is $f(f(x)) = x$

- (A) $\sqrt{2}$ (B) 1
(C) -1 (D) 2

[KEAM 2005, IIT JEE 2001]



79. *The function f defined by $f(x) = \frac{ax+b}{cx+d}$, where a, b, c and d are nonzero real numbers, has the properties $f(19) = 19$, $f(97) = 97$ and $f(f(x)) = x$ for all values except $-\frac{d}{c}$. Find the unique number that is not in the range of f .



Special Function II

$$f(x) = \sqrt{a^2 - x^2}$$



80. The range of the function given by the expression $\sqrt{16 - x^2}$ is

- (A) $[-4, 4]$
- (B) $[0, 4]$
- (C) $\{4, \sqrt{15}, \sqrt{12}, \sqrt{7}, 0\}$
- (D) $(0, 4]$

[NCERT EP]



81. Let $f : [-4, 2] \rightarrow \mathbb{R}$ be given by $f(x) = \sqrt{16 - x^2}$. Then the range of the function f is

- (A) $[0, 2]$
- (B) $[0, 2\sqrt{3}]$
- (C) $[0, 4]$
- (D) $[2\sqrt{3}, 4]$

[KEAM 2021]



Special Function III

$$f(x) = x + \frac{1}{x}$$



82. The function $f : \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as $f(x) = \frac{x}{1 + x^2}$, is

- (A) injective but not surjective
- (B) surjective but not injective
- (C) neither injective nor surjective
- (D) invertible

[JEE Main 2017]





83. Let $\mathbb{R}$ denote the set of all real numbers. Let $a_i, b_i \in \mathbb{R}$ for $i \in \{1, 2, 3\}$. Define the functions $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$, and $h : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = a_1 + 10x + a_2x^2 + a_3x^3 + x^4$$

$$g(x) = b_1 + 3x + b_2x^2 + b_3x^3 + x^4$$

$$h(x) = f(x+1) - g(x+2)$$

If $f(x) \neq g(x)$ for every $x \in \mathbb{R}$, then the coefficient of x^3 in $h(x)$ is

[JEE Advanced 2025]



Space for rough work



84. *Let the function $f : [0, 1] \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{4^x}{4^x + 2}$. Then the value of

$$f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + \cdots +$$

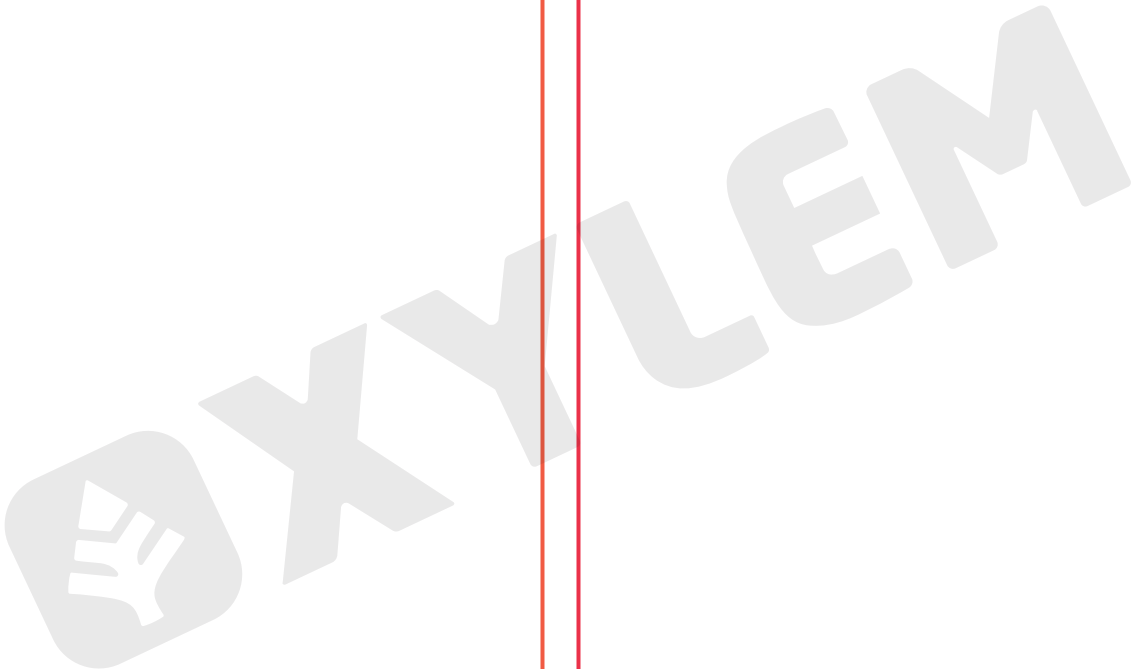
$$f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right) \text{ is } \dots\dots\dots$$



ANSWER KEY

(1) -	(29) (D)	(57) (C)
(2) -	(30) (D)	(58) (B)
(3) (A)	(31) (C)	(59) (C)
(4) (A)	(32) (D)	(60) 1
(5) 2^{16}	(33) 8	(61) 24
(6) (D)	(34) (B)	(62) (B), (C)
(7) 4	(35) (C)	(63) (D)
(8) (C)	(36) (A)	(64) (B)
(9) (D)	(37) 35	(65) (D)
(10) (B)	(38) (C)	(66) (D)
(11) (D)	(39) (C)	(67) (B)
(12) (C)	(40) (C)	(68) (B)
(13) (A)	(41) (A)	(69) (C)
(14) -	(42) (C)	(70) 8
(15) 7	(43) (C)	(71) (A)
(16) (D)	(44) (D)	(72) (B)
(17) (i) 16	(45) (B)	(73) (A)
(ii) 20	(46) (C)	(74) (D)
(18) 12	(47) (B)	(75) (A)
(19) $5^n - 2 \cdot 4^n + 3^n$	(48) (D)	(76) (A)
(20) (C)	(49) (B)	(77) (C)
(21) 170	(50) (A)	(78) (C)
(22) 4000	(51) (B)	(79) 58
(23) 3	(52) (A)	(80) (B)
(24) 0.8	(53) (C)	(81) (C)
(25) 126	(54) (D)	(82) (B)
(26) 365	(55) (B)	(83) -4
(27) 760	(56) (B)	(84) 19
(28) 570		

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